LECTURE 11: MATH1061/7861
TOPICS: ~~DIRECT PROOFS AND COUNTEREXAMPLES~~ Divisibility
DATE: WEDNESDAY MARCH 18, 2026

Learning objectives:

(1) Understand the definition of *'is divisible by'*.

(2) Understand and apply the Unique Factorisation Theorem.

$d \neq 0$ $\quad d \mid n$ means "d divides n" i.e. $\exists\, m \in \mathbb{Z}$ such that $n = md$.

There are many ways of saying this:

- n is divisible by d
- d is a factor of n
- n is a multiple of d.

Q1: Prove or disprove:

$$\forall a, b, m \in \mathbb{Z}, \quad m \mid (a+b) \longrightarrow m \mid (a-b).$$

$m \neq 0$

False.

$m = 3$, $b = 1$, $a = 2$.

Then $3 \mid 3 = (2+1.)$

$3 \nmid 1 = (2-1)$.

doesn't divide.

Q2: Prove or disprove:

$$\forall a, b, c \in \mathbb{Z}, \text{ if } c \neq 0 \text{ then } c \mid a \longrightarrow c \mid ab.$$

We prove the statement as follows:

$c \mid a$ means $a = kc$ for some $k \in \mathbb{Z}$

$$\Longrightarrow ab = (kc)b = c(kb)$$

Since $kb \in \mathbb{Z}$, this means $c \mid ab$.

Q3:

Write the prime factorisation of 360.

How many positive divisors does it have?

$$360 = 2 \times 180$$

$$= 2 \times 2 \times 90$$

$$= 2 \times 2 \times 2 \times 45$$

$$= 2 \times 2 \times 2 \times 3 \times 15$$

$$= 2 \times 2 \times 2 \times 3 \times 3 \times 5$$

$$\underline{= 2^3 \times 3^2 \times 5} \longleftarrow \text{Prime factorisation}$$

Suppose $n \mid 360 = 2^3 \times 3^2 \times 5$

The only possible prime divisors of n are $2, 3, 5$.

So $n = 2^{\square} 3^{\bigcirc} 5^{\triangle}$

$\square$ any number in $\{0, 1, 2, 3\} \leftarrow 4$

$\bigcirc$ any number in $\{0, 1, 2\} \leftarrow 3$

$\triangle$ any number in $\{0, 1\} \leftarrow 2$

$4 \times 3 \times 2$
$\|$
24
Total number of positive divisors

Q4:

Write the prime factorisation of 360^{30}.

How many positive divisors does it have?

$$\underbrace{360 \times 360 \times \cdots \times 360}_{30\text{ - times}}$$

$$360^{30} = \left(2^3 \times 3^2 \times 5\right)^{30}$$

$$= 2^{90} \times 3^{60} \times 5^{30}$$

So number of divisors is $\underline{91 \times 61 \times 31}$

$$= 172{,}081$$